

CMSC 104 Section 2

Spring 2026

Classwork #6

Strings and Lists

In-class date: April 6, 2026

Due date: before midnight, April 13, 2026

Assignment:

Part 1: String Manipulation

The assignment is to take one line of text that has been read in from a comma-separated value (.csv) file as a string, with elements separated by commas. Use “split” to break this string up into a list of values.

Use “isdigit()” on each element of the list to identify those that can be integers. For those elements, convert the element to an integer, determine if the integer value is less than 90. If so, add 10 to the integer value. Then convert the new value back to a string, because join only works with strings.

Then use “join” to re-assemble the values into a single new string where elements are separated by blank spaces.

Here is your string:

```
student = "Einstein,Albert,1A476B3,Graduate,94,86,97,92,105,17,54"
```

Your result should be the string

```
new_student = "Einstein Albert 1A476B3 Graduate 94 96 97 92 105 27  
64"
```

Part 2: Working With a One-dimensional List

Start with the same initial string as in part 1. That is,

```
student = "Einstein,Albert,1A476B3,Graduate,94,86,97,92,105,17,54"
```

Again, split the string on the commas, resulting in a list, called something like `student_list`.

Create a new, empty list, called `grade_list`.

Go through each element of `student_list`, and use `isdigit()` to identify each numeric element that can be converted into an integer. Append that integer value to the end of `grade_list`. Then, sum all of those integer values in `grade_list` to get a total grade for this student. If the student's grade is greater than 540 points, your program should print out something like

```
This student's grade is satisfactory for progressing to the next level.
```

If the student's grade is less than or equal to 540 points, your program should print out something like

```
This student must improve his/her skills before progressing to the next level.
```

Part 3: Working With Two-dimensional Lists

- a. First write code to generate a two-dimensional list containing the first 100 numbers, starting at 1 and ending at 100. These numbers should be arranged in order, with 10 rows each containing 10 numbers. In other words, your list should look something like this:

```
number_list = [  
    [1,2,3,4,5,6,7,8,9,10],  
    [11,12,13,14,15,16,17,18,19,20],  
    ...  
    [91,92,93,94,95,96,97,98,99,100]  
]
```

- b. Now write code to sum each of the 10 rows. Print out the sum of the numbers in each row, clearly labeled. Your output should look something like this:

```
The sum of the numbers from 1 to 10 is: 55  
The sum of the numbers from 11 to 20 is: 155  
...
```

```
The sum of the numbers from 91 to 100 is: 955
```

- c. Now write code to sum each of the 10 columns. That is, write code to sum $1+11+21+31+41+51+61+71+81+91$. Print out that answer, clearly labeled, similar to:

```
The sum of column 0 is 460.
```

Then sum the other columns in the same way, printing out the answers:

```
The sum of column 1 is 470
```

```
The sum of column 2 is 480
```

```
...
```

```
The sum of column 9 is 550
```

Reminder

Assignments are your own effort. Do not share your code..

Grading Rubric

Part 1 works correctly: 20 points

Part 2 works correctly: 40 points

Part 3 works correctly: 40 points

What to Submit

Your saved Python scripts. The easiest way to submit this assignment will be as three separate scripts. Please title your submission for Part 1 as `cw6_part1_(your_name).py` Please title your submissions for Parts 2 and 3 as `cw6_part2_(your_name).py` and `cw6_part_3_(your_name).py` respectively.